Algebra Review: Unit 6 — Exponents & Exponential Functions

Answer Key

Station 1: Properties of Exponents

1. x^7
2. $10a^5$
3. x^5
4. m^{10}
5. $27y^6$
6. $3a^3b^2$
7. $4x^7y^4$
8. $\dfrac{16x^4}{y^{12}}$
9. $-27x^6$
10. x^7

Station 2: Zero and Negative Exponents; Scientific Notation

1. 3
2. $\dfrac{1}{8}$
3. 9
4. $\dfrac{1}{x^5}$
5. $\dfrac{3b^4}{a^2}$
6. $\dfrac{x^6}{4}$
7. 4.5×10^6
8. 3.2×10^{-4}
9. 2×10^{10} $\quad 20 \times 10^9$
10. 4×10^4

Station 3: Exponential Functions: Evaluating and Identifying

1. 48
2. $\dfrac{5}{8}$
3. $\dfrac{1}{4}$
4. Growth; growth factor 1.5 *since $b > 1$*
5. Decay; decay factor 0.8 *since $0 < b < 1$*
6. y-intercept $(0, 10)$; growth factor 3
7. $y = 1,\ 3,\ 9,\ 27$
8. Yes (constant ratio 3); $y = 2 \cdot 3^x$
9. No: constant differences (linear, $y = 4x + 4$)
10. $y = 3 \cdot 2^x$ $\quad a = 3$*; growth factor* $\dfrac{6}{3} = 2$

Station 4: Exponential Growth and Decay Problems

1. $P = 500(1.04)^t;\ P(3) \approx 562$
2. $20{,}000(0.85)^4 \approx \$10{,}440.13$
3. $1000(1.05)^{10} \approx \$1{,}628.89$
4. $100 \cdot 2^6 = 6400$
5. $80\left(\dfrac{1}{2}\right)^3 = 10$ g $\quad \dfrac{24}{8} = 3$ *half-lives*
6. $P = 12{,}000(0.98)^t$
7. $12{,}000(0.98)^5 \approx 10{,}847$
8. (a) 7% growth (b) 8% decay
9. $3000\left(1 + \dfrac{0.04}{4}\right)^8 \approx \$3{,}248.57$
10. $2000 \cdot 2^3 = \$16{,}000$